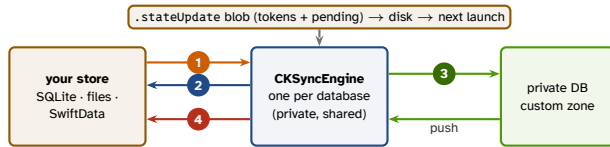


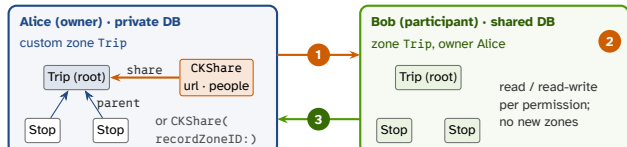
In one line: Three ways to sync: **CKSyncEngine** (iOS 17) — *your* store, the engine schedules push/fetch and you persist its state; **NSPersistentCloudKitContainer** (iOS 13) — Core Data *mirrors* itself into a zone, no sync code; **SwiftData** — the same mirror, private database only. **Sharing:** a CKShare in the owner's private custom zone covers one **record hierarchy** or the **whole zone**; an accepted share shows up in the participant's **shared database**, and their writes land in the *owner's* zone.

CKSyncEngine — you own the data, it owns the schedule



1 after each local write: `state.add(pendingRecordZoneChanges: [.saveRecord(id)])`
 2 engine asks `nextRecordZoneChangeBatch`; you hand over `CKRecords`
 3 sends ≤ 250 per request; creates its own `CKDatabaseSubscription`; retries rate-limit / network / zoneBusy itself
 4 handleEvent: `.fetchedRecordZoneChanges` → apply `.sentRecordZoneChanges` failures (incl. `serverRecordChanged: yours`) `.accountChange` → wipe

Sharing — the owner's zone, seen via the shared DB



1 share url out via `ShareLink` / `UICloudSharingController` (Messages, Mail)
 2 Bob taps it → app launched (`CKSharingSupported`) → `windowScene(_:userDidAcceptCloudKitShareWith:)` → `container.accept(metadata)` → the zone appears in his shared DB (`databaseChanges`)
 3 Bob's edits are written into **Alice's** zone and count against **her** quota

Which one?

	store	conflicts	sharing	public
raw ops / async	yours	yours	by hand	✓
CKSyncEngine	yours	yours	engine on shared DB	×
Core Data mirror	Core Data	the mirror	✓ iOS 15	✓ iOS 14
SwiftData	SwiftData	the mirror	×	×

CKSyncEngine — how it works (iOS 17)

- Create it early at launch with the saved `stateSerialization` — in any account state; it stays dormant until someone is signed in. Not for the public DB.
- Queue work: `.saveRecord(id)` / `.deleteRecord(id)`; zones via `add(pendingDatabaseChanges: [.saveZone(z)])`. Build batches with `RecordZoneChangeBatch(pendingChanges: recordProvider:)` — it stops at the limit.
- Schedule is *indeterminate* (battery, network): pull-to-refresh → `fetchChanges()`, “back up now” → `sendChanges()`; tests: `automaticallySync = false`.
- Keep each record's system fields (change tag) beside your row, or every save conflicts.

Example — the delegate

```
final class Sync: CKSyncEngineDelegate {
    var engine: CKSyncEngine!
    init(db: CKDatabase, saved: CKSyncEngine.State.Serialization?) {
        engine = CKSyncEngine(.init(database: db,
            stateSerialization: saved, delegate: self))
    }
    func edited(_ id: CKRecord.ID) { // after the local write
        engine.state.add(pendingRecordZoneChanges: [.saveRecord(id)])
    }
    func handleEvent(_ e: CKSyncEngine.Event, syncEngine: CKSyncEngine) async {
        switch e {
        case .stateUpdate(let u): store.persist(u.stateSerialization)
        case .fetchedRecordZoneChanges(let c): store.apply(c)
        case .sentRecordZoneChanges(let s): merge(s.failedRecordSaves)
        case .accountChange(let a): store.handle(a.changeType)
        default: break
        }
    }
    func nextRecordZoneChangeBatch(_ ctx: CKSyncEngine.SendChangesContext,
        syncEngine: CKSyncEngine) async -> CKSyncEngine.RecordZoneChangeBatch? {
        let todo = syncEngine.state.pendingRecordZoneChanges
            .filter { ctx.options.scope.contains($0) }
        return await .init(pendingChanges: todo) { store.record(for: $0) }
    }
}
```

Traps

- ! Not persisting `.stateUpdate` — every launch refetches everything.
- ! A pending `CKRecord.ID` built without its `zoneID` means the default zone — the engine keeps resending (*unverified*).
- ! `SwiftData` + an iCloud entitlement syncs *by default*; a .unique model then fails to load.
- ! No `CKSharingSupported` in `Info.plist` — the share link never opens your app.
- ! A child linked by your own reference field is *not* in a hierarchy share — set parent.

Core Data mirroring — NSPersistentCloudKitContainer

- Swap the class; capabilities `iCloud/CloudKit` + *Remote notifications*. Per store description: `cloudKitContainerOptions` (`containerIdentifier`, `databaseScope` `.private` `.public` `.shared`). Model configurations keep entities local.
- Runs off **persistent history**: set `NSPersistentHistoryTrackingKey` + the remote-change key, automatically `MergesChangesFromParent = true`, a merge policy.
- Model rules**: no unique constraints; attributes optional or defaulted; relationships optional *with an inverse* (records arrive out of order); no **Deny**; no ordered relationships (*unverified*).
- `initializeCloudKitSchema()` (DEBUG only) uploads sample records so Development has every field — then *Deploy*. Errors: `eventChangedNotification`.
- Sharing (iOS 15): 2nd store with `.shared` scope; `share(_:to:)`, `acceptShareInvitations(from:into:)`, `persistUpdatedShare`, `fetchShares`.

SwiftData + CloudKit

The same mirror. `ModelConfiguration(cloudKitDatabase:)`: `.automatic` (**default** — first container in the entitlements) `.private("iCloud.com.example.app")` `.none`. Same model rules. Private DB only: no public DB, no sharing API.

Sharing — CKShare

- Hierarchy**: `CKShare(rootRecord:)`; children by `setParent(_:)`; save share with the root. **Zone-wide** (iOS 15): `CKShare(recordZoneID:)`. A record is in **one** share, else `alreadyShared`.
- ≤ 100 participants, each with `iCloud`. `.readOnly` / `.readWrite`; roles `.owner` `.privateUser` (invited) `.publicUser` (`publicPermission > .none`: anyone with the link) `.administrator` (iOS 26, with `allowsAccessRequests`).
- UI: `ShareLink` + `CKShareTransferRepresentation` (iOS 16); `UICloudSharingController(share:container:)` manages people; `CKShare.SystemFieldKey.title` names the invite.
- Only the owner deletes the root; a participant deleting the share just leaves.

Remember

Engine: you store, it schedules. **Mirror:** it does both, you obey its model rules. **A share lives in the owner's zone.**